

Energy cable ampacity: Impact of seasonal and climate-related changes

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ABSTRACT

Climate change is influencing traditionally stable factors such as meteorological characteristics and soil conditions, impacting the planning process of electrical energy grids, especially energy cables. Supported by real-life data from the metropolitan region of Hamburg, this study examines the sensitivity of electric energy cables to seasonal and climate related changes, aiming to address inevitable future climate impacts. Using the thermal impedance model by the International Electrotechnical Commission, combined with 32 years of local soil and weather data, permissible current levels were calculated for a specific cable configuration. Comparisons with static boundaries reveal that shifts in environmental conditions can undermine the planning process, affecting maximum current limits and casting doubt on the current method's validity. Analysis shows that seasonal transitions significantly alter soil parameters within each annual cycle, causing up to a 10 % variation in energy transfer potential, depending on soil, cable, and regional specifics. Static standards also overestimate ampacity by up to 12 % for the studied region and timeframe. Climate change leads to shifting soil and weather conditions, causing unused energy transfer capacities, overestimations, and potential structural damage. As climate effects intensify, both seasonal and historical shifts are expected to have greater impacts, highlighting the limitations of the current static planning model without additional monitoring systems. As limited transmission capacities increasingly demand costly congestion management and equipment redundancies diminish, the need to optimize current resources and plan for a changing future becomes even more critical.

Nomenclature

Abbreviations				
AWC		Available water content		
DWD		German Meteorological Service		
IEC		International Electrotechnical Commission		
Symbol	Unit	Property	Description	Source
α	$\frac{\text{m}^2}{\text{s}}$	Thermal diffusivity	Ratio of thermal conductivity to volumetric heat capacity	Calculation with λ/C_v
δ_{soilFK}	–	Moisture level	Scaling factor for the useable field capacity	[1]
$\Delta\theta$	$^{\circ}\text{C}$	Permissible temperature rise	Maximum permissible temperature rise of the conductor above the ambient temperature	Formula (15) according to [40]

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Symbol	Unit	Property	Description	Source
$\theta_{a,\text{soil}}$	$^{\circ}\text{C}$	Ambient soil temperature	Ambient temperature of the soil	[37]
λ_{th}	$\frac{\text{W}}{\text{m} \cdot \text{K}}$	Thermal conductivity	Ability of a substance to transport thermal energy in the form of heat	Formula (7) according to [35]
λ_n	p.u.	Loss factors	Loss factors	Size depending on material and geometry
λ_1	p.u.	Loss factor for screen and sheath	Loss factors for screen and sheath	Calculation according to [40]
λ_2	p.u.	Loss factor for Armour	Loss factors for armor	Calculation according to [40]
ρ_d	$\frac{\text{t}}{\text{m}^3}$	Dry density of the soil	Density of the soil in the dried state	Material constant from DWD

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Symbol	Unit	Property	Description	Source
ρ_{th}	K m W ⁻¹	Specific thermal resistance of the soil	Specific resistance of the soil to the flow of heat	Formula (8)
AWC	p.u.	Available Water Content	Describes volumetric water content in the soil	Formula (11), formula and key figures from DWD [38]
C	J K ⁻¹ kg ⁻¹	Gravimetric heat capacity	Amount of energy (J,) required heat 1 kg (kg) of a given substance by 1 K (K)	Formula (1)
C _S	J K ⁻¹ kg ⁻¹	Specific heat capacity	Specific heat capacity of, for example, dry rock	Material constant
C _V	J m ⁻³ K ⁻¹	Volumetric heat capacity	Amount of energy (J) required heat a volume (m ³) by 1 K (K)	Formula (2)
C _W	J K ⁻¹ kg ⁻¹	Heat capacity of water	Heat capacity of water	Material constant
e	–	Porosity	Ratio of the volume of pores to the volume of the dried soil	Material constant according to [39]
I _{max}	–	Maximum current	Maximum current that can flow	Formula (5) according to [40]
m _d	kg	Dry mass	Mass of the dried soil	Material constant
m _w	kg	Water mass	Mass of the water	Material constant
n	–	Number of conductors	Number of conductors in a cable	Formulated assumption
nFK	–	Useable field capacity	Amount of water a soil can hold against gravity	[1]
R	Ω m ⁻¹	Material-dependent resistance	Resistance of the material	Calculation according to [40], material assumptions according to [41]
T _n	K m W ⁻¹	Material- and geometry-dependent temperature resistance	Temperature resistance of the material and geometry	Size depending on material and geometry
T ₁	K m W ⁻¹	Thermal resistance between conductor and dielectric	Resistance between conductor and dielectric against the heat flow of the cable	Calculated according to [40], material assumptions according to [41]
T ₂	K m W ⁻¹	Thermal resistance between sheath and armour	Resistance between sheath and armour against the heat flow of the cable	0, because no armour for assumed cable.
T ₃	K m W ⁻¹	Thermal resistance of external serving	Resistance of external serving against the heat flow of the cable	Calculated according to [40], material assumptions according to [41]
T ₄	K m W ⁻¹	Thermal resistance of the surrounding medium	Resistance of the surrounding medium against the heat flow of the cable	Formula (6) according to [40]
u	p.u.	Geometry factor	Dependent on depth and diameter of the cable	Determined according to [40], geometry according to the formulated assumptions [2]
V _d	m ³	Dry volume	Volume of the dried-out soil	[2]
V _p	m ³	Pore volume	Volume of the pores which is made up of water and air	[2]

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Symbol	Unit	Property	Description	Source
V _w	m ³	Water volume	Volume of the water in the pores of a soil	[2]
WG	wt. %	Weight percentage of water content	Proportion of the mass of water to the mass of the dried soil	Formula (14), derivation of formula (9)–(13) according to [39]
W _d	W m ⁻¹	Dielectric losses	Power loss per unit length	Calculated according to [40], material assumptions according to [41]

1. Introduction

Climate change has wide-ranging impacts on various aspects of modern life. The overall rise in global temperatures has a number of detrimental consequences: deviations in soil temperature, declining groundwater levels, and soil drying are just a few examples [3]. In addition to these inevitable absolute changes, meteorological characteristics, soil conditions, and temperatures are becoming increasingly volatile [4]. Some regions are experiencing the combination of these shifts, transitioning from a previously moderate and stable environment to one that is progressively hotter and drier, with more pronounced extremes. These circumstances present significant challenges that require careful attention.

Processes and structures with extended life cycles and long planning periods are notably impacted. These, in particular, must feature resilience to climatic variations and the ability to adapt to volatile environments. Electric energy cables have a regular life cycle of at least 40 years and their grid integration and energy transport capacity have to be designed for at least that period [5,6]. As part of the critical infrastructure, energy grids and their cables must be durable and function flawlessly. Permissible planning and operational errors depend on factors such as the voltage level, the number of stakeholders affected by a fault, and the time required for recovery from a blackout. To ensure reliability, cable manufacturers typically plan and calculate energy cable connections based on worst-case scenarios during the initial grid integration phase [7,8]. It is generally accepted that unused energy transfer capacities exist, and the maximum permissible current cannot be exceeded. This approach has been deemed accurate enough due to stable environmental conditions. However, given the current climate and the increasingly significant shifts in weather patterns, this assumption must be reevaluated.

There have been numerous studies that calculate the ampacity for various cables, voltage levels, and installations. These studies have employed methods such as the finite element method or thermal impedance approaches, such as the Neher-McGrath method, the IEC 60287 standard or a derived national version [9–14]. A summary of emerging technologies from the last years and their methodologies showed that both concepts yield similar results for single line cables [15, 16]. Additionally, efforts have been made to understand the aging behavior of cable insulation and any fuses used, as these components often limit the lifespan of the cables [17–19]. Alternative approaches involve dynamically expanding the energy transfer capacity through the monitoring of operating temperatures [20]. However, these methods typically require additional components such as fiber optic cables or distributed temperature sensors, or they cannot be used during grid operation, limiting their applicability [21,22]. Harmonics were found to be an additional external influence causing excess heat and losses which in turn affect the ampacity [23]. Different methodologies for determining the permissible ampacity have been analyzed, all of which are rooted in the thermal impedance model by Neher-McGrath [24]. Another study identified the conductor size, native soil resistivity, and grounding layout as the primary factors influencing the ampacity [25]. Some studies conduct sensitivity analyses on the parameters of these

models, but most do not correlate these operating variables with soil parameters such as saturation, temperature and thermal conductivity [26]. Most that focus on the ampacity operating limit and the effects of controlled environments, do not incorporate real measurement data to draw additional conclusions [27]. One research paper demonstrated that daily soil moisture variations significantly reduce the current-carrying capacity of buried cables in semi-arid conditions, underscoring the importance of soil moisture monitoring to optimize cable performance and design [28]. This work utilizes the IEC 60287 thermal impedance model even though a finite element method would be equally suitable. The model used herein distinguishes itself by deriving the input parameters empirically from long term soil measurements from the metropolitan region of Hamburg.

Studies have already established a connection between climate events and the ampacity rating of transmission lines, highlighting that increasing temperatures lead to reduced transmission capacities [28–30]. They distinguish themselves by the regions analyzed and their scope besides the current rating analysis. This research often includes changes in energy consumption, population trends, and other environmental hazards which all contribute to an adverse transmission capacity trend. A study investigating the electric grid of California found that overhead line capacities are reduced by an average of 7.5 percent [29]. Other research in related fields has assessed climatic impacts on energy consumption rates of buildings and residential demand areas [29,30]. Several other studies have linked weather events, such as rainfall, to changing soil conditions and resulting variations in ampacity, though they did not place these events in the broader context of climate or seasonal patterns [31]. There are some studies that evaluate the risks of climate change on electrical infrastructure, also with a focus on submarine cables [32]. However, they do not focus on the ampacity rating. Others link climate conditions, particularly seasonal weather variations, to failure rates of underground cable joints in Dubai's distribution networks, using predictive modelling to improve maintenance planning [33]. To the best of the authors' knowledge, there is no study that analyzes electric cable ampacity in the context of climatic changes. This work further stands out methodologically from existing research by combining a sensitivity analysis of the model parameters with a regression of real soil data. Relevant local trends were derived by analysing real data obtained from measuring stations. These findings highlight both unused transmission potentials and overestimations. The ampacity results for electric energy cables in the metropolitan region of Hamburg provide unique insights into the association between ampacity and climate-related and seasonal changes.

Due to the complexity of this multi discipline topic, the literature research is summed up by a comparative analysis of methodologically and topically related research. Table 1 provides an overview of this research's distinguishable contributions and how the reviewed literature may be compared. The methods used in this study include a literature review and comparative analysis, sensitivity analysis, empirical analysis, and regression analysis. This work addresses key aspects such as the use of real-world soil measurements, application of the IEC thermal impedance model, focus on electric power cables, analysis of climate-related impacts, and a case study focused on a specific region.

The central research question examines whether the currently used static methods for determining the ampacity of electric energy cables remain suitable in the context of climate-related changes. Novel contributions include the case study with over 30 years of real-world soil data and the derivation of historical ampacity trends. This trend, in the context of climate change, has already been demonstrated for overhead lines but not for underground cables [34–36]. With this correlation made and a literature review conducted, it became evident that no universally applicable methods exist for dynamically assessing ampacity limits of energy cables. Therefore, criteria for such a method have been outlined.

This research begins by assessing the ground and soil conditions in the metropolitan region of Hamburg and contextualizing them with climate models and trends. It then proceeds to review the existing procedures for planning and calculating the parameters of electric energy cables. A comprehensive analysis is conducted to identify the parameters affected by climate change, followed by a sensitivity analysis. Real ground and soil data from the area spanning from the year 1991 to the present are utilized to apply the calculation model under actual climatic conditions. The work presents a case study where a specific cable and its non-climate-related parameters are predefined to calculate the maximum current for each measuring station within Hamburg's metropolitan area. By comparing the results of two different measuring stations over a 32-year time period, the study highlights the deviation from the conventional planning process. This establishes an indicator of the quantitative impact of influential environmental and seasonal changes including the effects of climate change. It is further explored whether the current methodology for determining an energy cable's maximum current, and consequently its energy transfer capacity, remains suitable given the observed deviations. Additionally, alternative solutions are discussed to address the challenges posed by changing environmental conditions.

Table 1
Comparative analysis overview.

Research item	Methods used ^a	Key distinguishing aspects				
		soil measurements	IEC ampacity model	electric energy cables	climate related changes	case study of a region/country
This work	comparative analysis, sensitivity analysis, empirical analysis, and regression analysis	X	X	X	X	X
[3]	literature review, model integration				X	
[5,6,10,11]	empirical analysis or laboratory experiment	X		X		
[9]	literature review, empirical analysis, laboratory experiment	X	X	X		
[12,14,15,19,25,26]	sensitivity analysis		X	X		
[16,17]	literature review, model integration		X	X		
[20,21]	literature review, empirical analysis			X		
[22,23]	model integration		X	X		
[28]	literature review		X	X		
[27]	field study	X		X		
[28–30]	literature review, model integration				X	X
[29,30]	literature review, model integration				X	
[32]	empirical analysis, model integration			X	X	X
[28]	comparative analysis, sensitivity analysis, empirical analysis, model integration	X	X	X		X

^a Models used that are relevant for the distinctions of this contribution.

2. Ground and soil analysis of the metropolitan region of Hamburg

Climate change has both direct and reciprocal effects on soils. The climate influences the physical, chemical, and biological processes occurring within the soil. Soil formation is a time-consuming process, whereas biological processes operate on a much faster timescale. Temperature and water availability play a crucial role in these processes. Conversely, soils also influence the climate through the emission of climate-active trace gases and the storage of rainwater. Additionally, the climate naturally affects the mechanical and physical properties of soils, which are important for the settlement and utilization of areas. Consequently, altered soil characteristics resulting from changing climatic conditions are a significant influencing factor for soil-based technical infrastructure such as energy networks [37].

Considering the metropolitan region of Hamburg, two fundamental areas can be defined with distinct soil types: Along the coasts and in areas near the Elbe River, marsh and alluvial soils prevail. Primarily nutrient-rich clay soils can be found here, formed from marine or fluvial deposits. In the eastern region of the Elbe, areas with clayey to loamy alluvial soils are more prevalent. In the inner land areas, within the glacially shaped geest landscape, soils of the undulating-flat countryside, broad river valleys, and lowlands can be found. Here, predominantly clayey-sandy and relatively nutrient-poor soils dominate. By contrast, the central urban areas are characterized by anthropogenically strongly overformed and sealed soils [38].

To enable a long-term, comparable, and standardized evaluation of the impacts of climate change on soils and subsequently on buried cables, this study relied on data provided by the German Meteorological Service (DWD). The DWD offers data from approximately 485 soil measurement stations, 30 of them in the metropolitan region of Hamburg, which provide a repository of monthly recorded values for the past 32 years. For this case study, the DWD standard soil composition was employed, consisting of 30% sand, 55% silt, and 15% clay, with a dry density of 1.2 t/m³. It has a wilting point of 12%, a natural field capacity of 24%, and a maximum saturation level of 37% [5,6]. These reference parameters are reasonable estimations, but the actual values may vary for specific circumstances, places, and times. The following sections will explain how climate change impacts the metropolitan region of Hamburg and how it will affect buried cables. They will focus on the most important parameters for belowground cables and the expected climate changes in the region over the next 30 years.

2.1. Climate and its models in general and in Hamburg

The World Meteorological Organization defines climate as a “synthesis of weather over a period long enough to allow its statistical properties to be determined” [39]. To consider climatic changes, international comparison periods of 30 years have been agreed upon. For the presentation of climatic changes, the observed and projected changes are usually considered in relation to the reference period (e.g., from 1961 to 1990) [40]. Climatic conditions on Earth are variable. On the one hand, the components of the climate system, such as the atmosphere, the biosphere, and the hydrosphere, interact in many ways. Also, there are natural factors, such as solar activity, the Earth’s orbit, and volcanic eruptions, that can affect the climate. On the other hand, human activities have a significant influence on the climate system, particularly through changes in land use and the emission of greenhouse gases. These changes have significant impacts on the concentration of greenhouse gases in the atmosphere. The consequences are referred to as human-induced climate change and are the focus of this paper. The term ‘climate change’ is always understood to mean climatic changes caused by anthropogenic activities.

Computer-based climate models are used to simulate the complex interactions in the climate system and the resulting climatic conditions in a specific geographical area. These climate models combine the

results of various sub-models for individual processes into a larger model. The world’s most diverse climate models have been created independently with various assumptions, distinct sub-models and calculation methods, and with diverse levels of granularity. To see if the models are plausible and to improve them, the model results are usually compared to the actual climatic conditions in a reference period. To project the possible climatic conditions in the future, the representative concentration pathway (RCP) scenarios are used as input for the climate models [41]. As a result, numerous model results are available for each RCP scenario for different time periods, which allow conclusions to be drawn about the range and probability of the projected changes. The DWD has issued the Climate Report Hamburg, which shows how the Northwest German Lowlands, which include Hamburg, will change in 2021–2050 and 2071–2100 [40].

2.2. Relevant climate effects preceding the change in the soil’s thermal conductivity

The thermal properties of the soil can be described by three variables [42]. They essentially determine the suitability of the soil for the extraction of geothermal energy:

- **Thermal conductivity** λ_{th} in W/(m·K): Ability of a substance to transport thermal energy in the form of heat.
- **Specific heat capacity** C in J/(kg·K): C_V in J/(m³·K): Amount of energy (J, Joule) required to heat 1 kg (kg) of a given substance by 1 K (K)
- **Thermal diffusivity** α in m²/s: Quotient of thermal conductivity and volumetric heat capacity ($\alpha = \lambda/C_V$), a measure of the penetration speed and the attenuation of a heat flow.

For these three properties of soils, effects such as soil moisture, soil temperature, and type have a relevant influence. One example is the thermal conductivity. Moist soils have higher heat capacities than dry soils [43]. As a result, more energy must be supplied to them in order to cause a temperature increase. In the case of thermal use (energy extraction), however, more energy can also be extracted until the temperature changes. Additionally, C describes the same impact of soil moisture, temperature, and type.

$$C = \frac{(100 \cdot C_s + C_W \cdot WG)}{100 + WG} \quad (1)$$

$$C_V = \rho_d \cdot C \quad (2)$$

C_s Specific heat in $\frac{J}{(K \cdot kg)}$

C_V Volumetric heat capacity in $\frac{J}{(K \cdot m^3)}$

C_W Heat capacity of water in $\frac{J}{(K \cdot kg)}$

WG gravimetric water proportion in wt. %

ρ_d dry bulk density of the soil in t/m³

In addition to the water content, the temperature also has an influence on heat capacity. The heat capacity of dry rock for example with elevating temperatures [43]. The interaction of soil moisture, temperature, and type is therefore relevant for further consideration. Since the soil type can usually be assumed to be static, soil moisture and soil temperature are most central. Both depend on environmental conditions, including ambient temperature and precipitation. In addition, groundwater level, vegetation, buildings, and other factors are also identified as properties, albeit properties that usually follow the climate development with a considerable delay [43,44]. Therefore, this study is limited to ambient temperature and precipitation and their potential development in the area of study.

2.3. Development of temperature and precipitation in the Hamburg metropolitan area

In this section, the development of the climate parameters temperature and precipitation, which are taken from [40], will be explained as influencing climate factors. Since a variety of climate models, approaches, and scenarios were considered in [40], the following paragraphs only serve as an adumbrated summary of the possible developments. The time frame considered is the more distant long-term scenario (2071–2100).

- **Temperature:** The air temperature at a height of 2 m above the ground is examined. For Hamburg, the mean annual average temperature in the most recent reference period (1991–2020) is 9.8 °C. Since pre-industrial times (1881–1910), the temperature in Hamburg has increased by 1.7 °C. The average annual temperature will increase by an amount ranging from 0.5 to 3 °C by the end of the century (2071–2100). This trend will continue in the future [40].
- **Precipitation:** The study of rainfall, similar to the study of the temperature, emphasizes the need for the examination of the average values and patterns of both the annual rainfall and the seasonal rainfall totals. Precipitation behavior is closely connected to atmospheric circulation, which is responsible for bringing in air masses from different directions and with different relative humidity values, temperatures, and stratification stability. The distribution of precipitation in the metropolitan region of Hamburg is primarily determined by the geographical location. The total amount of precipitation per year shows an increasing trend, as does the temperature. In 1961–1990, an average precipitation amount of 750 l/m² was recorded. In 1991–2020 this amount was 773 l/m². A breakdown to seasonal values reveals that the increase is greatest in winter. The precipitation distribution has changed in such a way that the share of winter precipitation has increased significantly [45]. In addition to the change in the annual average, the precipitation extremes are also changing. Since 1951, the number of days in Hamburg with at least 10 mm of rain in 24 h has increased by 3.4 days per year [45]. Conversely, the number of rainy days is decreasing, rendering droughts a problem of increasing severity. The years 2018–2020 in particular have shown the problems this may cause, especially regarding the change in soil properties due to erosions, floods, and the resulting change in thermal resistivity. Climate projections predict that this trend will continue: annual precipitation will increase but unevenly so over the seasons leading to disproportionately more rain during winter while rendering the summer drier. An increase in the frequency and intensity of heavy rainfall events must also be expected [40].

In summary, the general trend up to the year 2100 points to an increase in extreme and average temperatures as well as an increase in extreme precipitation and a shift of precipitation into winter.

3. Review of the current planning process of energy cables and their energy transfer capacity

The planning process of energy cables is embedded into the encompassing grid design. The energy transfer need between two grid connection points is calculated with grid simulation tools to account for all generation and loads, other transfer lines, load management systems, different voltage levels, and redundancies required to ensure resilient and economical operation. Once the transfer power is determined and the necessity for a cable is established, a proper cable is chosen by adhering to the following criteria:

- The dielectric medium has to withstand the operating voltage phase-phase U_{Δ} and a permissible voltage deviation ΔU_{Δ} .

- The length and load-dependent voltage drop over the cable must not violate the permissible voltage deviation ΔU_{Δ} . The most critical local voltage deviation must not exceed 10% of the nominal voltage within the European grid operated by the European Network of Transmission System Operators for Electricity.
- The resulting apparent power S has to be sufficiently high to cover all simulation cases, see eq. (3).
- The maximum rated current I_r has to be below the manufacturer's limit during all operation.
- Economic factors, availability, and route specific auxiliary factors e. g., the bending radius or maximum diameter of the cable, have to be considered as well.

$$S = \sqrt{3} \cdot U_{\Delta} \cdot I_r \quad (3)$$

While most of the electrical criteria are predetermined and part of broader canonical regulation, the rated current I_r is set by the cable manufacturer for the sole purpose of limiting the actual operating temperature of the cable to a certain limit. This limit is derived from material properties of the cable or a permissible rise in the ambient temperature. Therefore, the permissible maximum current $I_{\max}$ is the operational variable and is therefore examined more closely in this paper.

Current heat losses and external influences such as the ambient soil temperature and additional heat sources such as other power cables or central heating pipes contribute to the final operating temperature of energy cables. Internal properties such as materials, geometric relations and electric operating parameters, and certain external ones, e.g., the way the cable is laid and how deep it is buried, are used to model an approximation of the stationary operating temperature for a given current I_r . The maximum current $I_{\max}$ ensures that no thermal failures occur and that the permissible rise in ambient temperature is not exceeded. The International Electrotechnical Commission (IEC) offers an analytic approach with following correlations (4) and (5) [8]:

$$I_{\max} = f(\Delta\theta, T_n, W_d, R, \lambda_n) \quad (4)$$

$\Delta\theta$ permissible temperature rise of the conductor above ambient temperature in °C

T_n material and geometry-temperature resistance in $\frac{(K \cdot m)}{W}$

W_d dielectric losses in $\frac{W}{m}$

R material in $\frac{\Omega}{m}$

λ_n loss factors in p.u.

n number of conductors in a cable

$$I_{\max} = \sqrt{\frac{\Delta\theta - W_d \cdot [0.5 \cdot T_1 + n \cdot (T_2 + T_3 + T_4)]}{R \cdot T_1 + n \cdot R \cdot (1 + \lambda_1) \cdot T_2 + n \cdot R \cdot (1 + \lambda_1 + \lambda_2) \cdot (T_3 + T_4)}} \quad (5)$$

This approach is necessary because there is no reliable way to measure the operating temperature of the cable directly. The installation of distributed temperature sensors is often not possible once a cable is already placed and rarely economical for new ones. There are other solutions available, but they all require a modified cable structure. An example is the integration of a fiber optic cable into the cable's screen to measure the temperature-dependent distortion of the light wave. Neither additional temperature sensors nor modified cable structures are practical solutions for a large-scale grid integration [46–49].

The permissible maximum current $I_{\max}$, according to eq. (5), is a limit that is conventionally understood as conservative and adherence to which will result in the protection from thermal failures and inadmissible increases of the surrounding temperature. This means that worst case scenarios must always be taken into account when parametrizing the equation, leaving ample potential unused during normal or optimal conditions. It also means that the worst-case scenario has to be known and remain constant during the life cycle of the cable. Seasonal changes inherently vary the operating conditions and climate change will exacerbate the currently assumed worst cases. Thus, climate change's effects

on the parametrization of the model are investigated further by applying local soil data from the DWD measuring stations to the IEC method.

3.1. Climate-dependent parameters

Some parameters considered in eq. (5) are directly connected to the ambient conditions of the soil and the surrounding area. The values for these parameters are estimates of the actual surrounding conditions and are subject to external change. The assumption of homogeneous ground structures is already a significant simplification, which leads to deviations and inaccuracies in the planning process of energy cables. The planning method includes the following climate-related parameters [8]:

- T_4 thermal resistance of surrounding medium in $\text{K} \cdot \frac{\text{m}}{\text{W}}$
(ratio of cable surface temperature rise above ambient to the losses per unit length)
- $\theta_{a,\text{soil}}$ ambient temperature in $^{\circ}\text{C}$
- $\Delta\theta$ permissible temperature rise of the conductor above ambient temperature in $^{\circ}\text{C}$
- ρ_{th} thermal resistivity of the soil in $\text{K} \cdot \frac{\text{m}}{\text{W}}$

u geometric factor dependent on the installation depth and diameter of the cable in p.u.

Eq. (6) correlates the thermal resistance of the surrounding medium and the thermal resistivity [50]:

$$T_4 = \frac{1}{2\pi} \cdot \rho_{\text{th}} \cdot \ln(u + \sqrt{u^2 - 1}). \quad (6)$$

Especially ambient temperature θ_a and thermal resistivity $\rho_{d,w}$ of the soil are not only subjected to local variations due to vegetation, infrastructure, moisture, and different soil types but also to regular seasonal changes and climate induced irregularities [51,52].

The German Federal Institute for Geosciences and Natural Resources offers a correlation, see eqs. (7) and (8), between the thermal conductivity λ_{th} and the soil moisture level for specific soil types as well as between the former and the thermal resistivity ρ_{th} [1,43,53].

$$\lambda_{\text{th}} = 0.1442 \cdot (0.9 \cdot \lg(WG) - 0.2) \cdot 10^{0.6243 \cdot \rho_d} \quad (7)$$

$$\rho_{\text{th}} = \frac{1}{\lambda_{\text{th}}} \quad (8)$$

This work presents a proposal for deriving the required parameter WG from the DWD input data $\delta_{\% \text{nfK}}$. Similar model connections have been established, but variations in input parameters necessitated deriving a new correlation approach [54]. WG can be expressed as the fraction of the mass of water to the mass of the dried-out soil, see eq. (9) [1,55]. The dry density ρ_d of the soil and the density of water ρ_w connect these parameters to their respective volumes V_d and V_w . The moisture levels $\delta_{\% \text{nfK}}$, as a percentage of the useable field capacity, was used with a maximum useable field capacity (nfK) of 24 % and a wilting point (WP) of 13 % to calculate the available water content (AWC) [55,56]. The pore number e is a soil-specific parameter to describe the volumetric relation of the pores to the dried-out soil with an approximate value for e of 0.4 for the soil investigated [2,16].

$$WG = \frac{m_w}{m_d} = \left(\frac{\rho_w}{\rho_d} \right) \cdot \left(\frac{V_w}{V_d} \right) \quad (9)$$

$$WG = \frac{\rho_w}{\rho_d} \cdot \left(\frac{V_w}{V_p} \right) \cdot \left(\frac{V_p}{V_d} \right) \quad (10)$$

$$AWC = \left(\frac{V_w}{V_p} \right) = \frac{\delta_{\% \text{nfK}}}{100} \cdot \text{nfK} + \text{WP} \quad (11)$$

$$e = \left(\frac{V_p}{V_d} \right) \approx 0.4 \quad (12)$$

$$WG = \frac{\rho_w}{\rho_d} \cdot AWC \cdot e \quad (13)$$

$$WG\% = \frac{\rho_w}{\rho_d} \cdot AWC \cdot e \cdot 100 \quad (14)$$

The permissible temperature rise $\Delta\theta$ is calculated by using the surrounding temperature of the soil and maximum temperature of the conductor, see eq. (15) [8].

$$\Delta\theta = \theta_{\text{conductor,max}} - \theta_{a,\text{soil}} \quad (15)$$

3.2. Sensitivity of the planning process towards changing parameters

The following analysis shows how the parameters from section 3.1, the ambient temperature θ_a , and the thermal resistivity $\rho_{d,w}$, affect the resulting maximum current I_{max} for the IEC model. The example is based on the study case that will be presented in section 4. The ambient temperature was varied between 0°C and 45°C . The thermal resistivity was varied between $0.1 \text{ K} \cdot \text{m/W}$ and $4.0 \text{ K} \cdot \text{m/W}$ [57].

The basis for the sensitivity analysis and the case study was assumed to be a three-phase system consisting of three single-core medium voltage cables of type N2XS2Y $1 \times 35/16 \text{ 6/10 kV}$ laid in trefoil. The cables have an external diameter of 24 cm and are laid in the ground at a depth of 50 cm. A grid frequency of 50 Hz is assumed, along with constant capacity, cable geometry, cable distances, and loading. The material constants were taken from the datasheet [58]. The cable has no armor, and hence the derived values are zero.

Fig. 1 shows, firstly, a linear correlation of the maximum operating current to the soil temperature and, secondly, an exponential relationship of the former to the thermal conductivity of the soil. Even today's extreme scenarios for cold and wet soils, with an ambient temperature of 0°C and a thermal resistivity of $1 \text{ K} \cdot \text{m/W}$, and for hot and dry soils, with an ambient temperature of 20°C and a thermal resistivity of $2.5 \text{ K} \cdot \text{m/W}$, cause a major difference in the permissible current I_{max} [57]. The obtained values for I_{max} for the optimal and the worst-case scenario are 245 A and 148 A, respectively. This shows the magnitude of the influence of the soil factors, which are seasonally changing and impacted by climate change, and their impact on the estimation of a permissible maximum current in energy cables. The quality of their estimation can introduce a relative error of up to almost 60%. The approximation error will increase even further with more impactful climatic changes. The reliability of a forecast for a cable's transfer capacity diminishes when the limits of the soil parameters change or expand. The simple fact of climate change-induced alterations in the soil parameters introduces further inaccuracies.

4. Case study of a default energy cable and its results

The case study was conducted to emphasize the impact of actual local soil and weather parameters on the maximum operating current within the metropolitan region of Hamburg. The IEC model from section 3 was used in a fixed setup to calculate the permissible current limit with eq. (5) in two ways: first, using reference parameters from the standard and, second, using data from soil measurements. A three-phase system with three common one-conductor medium voltage cables of the type N2XS2Y $1 \times 35/16 \text{ 6/10 kV}$ was chosen and assumed to be buried in the trefoil position in 50 cm depth. The cables are grounded on both ends, and the system is loaded symmetrically and operated at 50 Hz [23,25]. The surrounding soil is the DWD-standard soil described in section 2. The cable manufacturer declares a maximum operation current of 187 A for buried cables under these specified conditions, see eq. (17) [58]. This upper limit is expected to suffice under any surrounding conditions and for all soil types.

A few simplifications regarding the cable placement were employed to apply the IEC model properly. The cables' positions to one another are

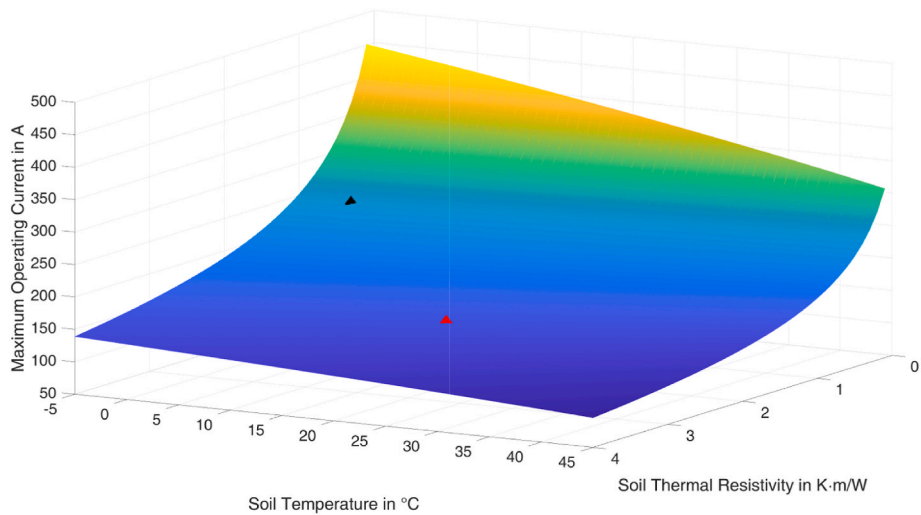


Fig. 1. Impact of soil properties on the maximum operating current of an energy cable. Black and red markers indicate 245 A at (0 °C, 1 K-m/W) and 148 A at (20 °C, 2.5 K-m/W), respectively. (For interpretation of the references to color in this figure legend, the reader is referred to the Web version of this article.)

constant and the thermal expansion is negligible. The soil type, its composition, water content, and temperature remain constant within the examined area which results in stationary conditions for a selected time interval.

Local monthly soil data from the German Meteorological Service (DWD) of 30 measuring stations was obtained for the timespan of 1991 until 2022. As described in section 2, a DWD-standard soil of 30% sand, 55% silt, and 15% clay, with a dry density of 1.2 t/m³ was assumed. The pore number e is a reference value for this soil type, which varies between 0.35 and 0.45 [2]. The data sets TS50 and BF50 were used to

describe the conditions in 40–50 cm depth. Cables are usually buried at least 60 cm into the ground. This results in a 3% error when calculating the external thermal resistance T_4 , which was not further investigated. BF50 describes volumetric moistures, which scale the useable field capacity (nFK) and are used to calculate the gravimetric available water content (AWC). Definitions of the water content (WG) and the AWC vary in the literature. Eqs. (9)–(14) define the key figures in the context of this case study. The metropolitan area of Hamburg and the 30 measuring stations of the DWD are shown in Fig. 2 [55]. The integrated table further indicates the difference between the results of the IEC model for

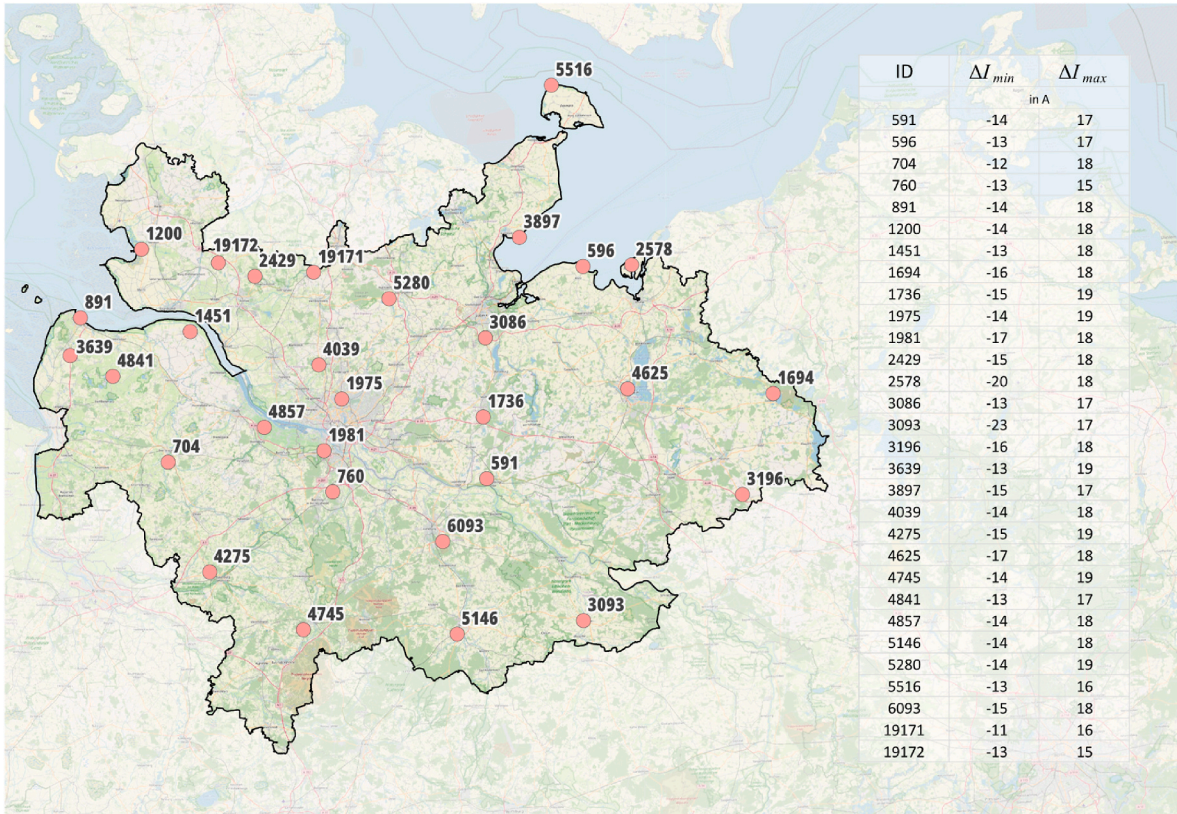


Fig. 2. Hamburg's metropolitan region, its measuring stations from the German Metrological Service (DWD), and the derived minimum and maximal values of the ampacity.

the permissible current when local soil data is applied and when the generic reference values of the standard are used. The difference is derived from eqs. (16) and (17). These values are extremes for the entire time range (1991–2022). They represent the absolute maximum and minimum energy transfer potentials, expressed as a possible increase in operating current.

$$[\Delta I_{\min} \Delta I_{\max}] = [\min(I_{\text{station}}) - I_{\text{IEC,ref}} \quad \max(I_{\text{station}}) - I_{\text{IEC,ref}}] \quad (16)$$

$$I_{\text{IEC,ref}} = 187\text{A} \quad (17)$$

The soil parameters relevant for calculating the ampacity were presented in section 3.1 and the effects of their change in section 3.2. While a uniform soil composition is assumed in this study, the soil temperature and moisture of the individual stations vary significantly. This variation can be caused locally or due to the micro-geographical location. Locally, the vertically adjacent soil layers have a particular effect. If, for example, the subsoil has a very high clay content or is solid rock, this can lead to alluvial deposits and increase the water content [59]. If the overlying soil layer is similarly impermeable or obstructed, this can reduce the local water content [59]. However, this distinction is not appropriate for the scope of the study, as these are generally local phenomena that do not directly represent climatic and seasonal effects. In addition to the lack of a data basis and the issue that many other influencing factors are unknown or undefined, comparability between the measuring stations would also be lost. However, the situation is different with the micro-geographical location. Although other notable distinguishing features exist, such as the degree of urbanization, altitude, natural vegetation, and the agricultural use of the top soil, stations 3093 and 3639 were selected for further investigation as examples because they capture the full range of the oceanic influence on the soil parameters. This selection criterion is exemplary and serves to illustrate the differences within the region. A more extensive study is outlined in the discussion in section 5.4, as this is beyond the scope of this paper. Station 3093 is close to the small town of Lüchow which is far away from the Baltic and North Sea. It is therefore less influenced by these water bodies and their effects on ground water levels and thermal stability. Station 3639 is close to Cuxhaven and only 9 km away from the North Sea and thus highly influenced by that water body.

Firstly, the seasonal changes over the last 10 years are shown in Fig. 3 for these two stations. The moisture levels of station 3639 are less volatile, and soil temperatures are slightly more moderate than the values of station 3093. These are expected phenomena due to the former station's proximity to the sea. The derived ampacity for the two stations is shown together with the common IEC standard reference for that cable type. It can be observed that both stations show a higher

permissible ampacity than the reference for the winter seasons. In the summer seasons, however, measurements from both stations lead to ampacities below the IEC standard. The noticeable decline in soil moisture of station 3039 in 2019 led to a minor decline in thermal resistivity and ampacity.

In a highly reproducible pattern, January is the coldest month and July the warmest, which is why they were chosen for the following analysis. The comparability of the year-to-year development was prioritized over the display of the yearly absolute maximum and minimum values. Fig. 4 shows moisture level, temperature, and ampacity over the entire 32-year time range for January and July for both measuring stations.

Both stations show moderate temperature fluctuations. Station 3093 recorded a moisture level that is more volatile, especially during winter. It also shows a significant drop of the moisture level in 2019. Station 3639 measured a moisture level that barely varies during the winter and shows slightly more volatility during summer. On average, the ampacity of station 3639 is on average 8% higher than the reference during winter and about 4% lower during summer. According to the examined data, the ampacity during the summer month always lies below the IEC reference value, whereas the ampacity of the winter months always superceeds this limit. On average, the ampacity of station 3093 is 7% higher than the reference during the winter and about 5% lower during the summer.

Secondly, a linear regression model allows an interpretation of temperature, moisture, and ampacity trends over the period under investigation. Fig. 5 displays the difference of the characteristic time-dependent data and the respective reference value from 1991 for measuring stations 3093 and 3639 for each month. Development trends were derived by comparing the individual months over the years. Soil temperatures generally rise with a minimum increase of 0.76 °C in May and a maximum increase of 2.73 °C in November for station 3093 and a minimum increase of 0.41 °C in May and a maximum increase of 2.64 °C in November for station 3639. The soil moisture levels decrease with a minimum reduction of 2.49% of the AWC in February and a maximum reduction of 10.42% of the AWC in December for station 3093. However, the trend of the moisture levels of station 3639 is not uniform. Decreases of 3.67% of the AWC in September and December get counteracted by increases of about 2.5% in February, June, and October. The proximity to the ocean mainly drives this trend. However, the resulting ampacity decreases for both stations, with a minimum reduction of 2.3 A in February and a maximum reduction of 5.4 A in November for station 3093 and a minimum reduction of 0.1 A in May and a maximum reduction of 3.3 A in December for station 3639. Applying equation (3), transfer power reductions for the default cable system at station 3093

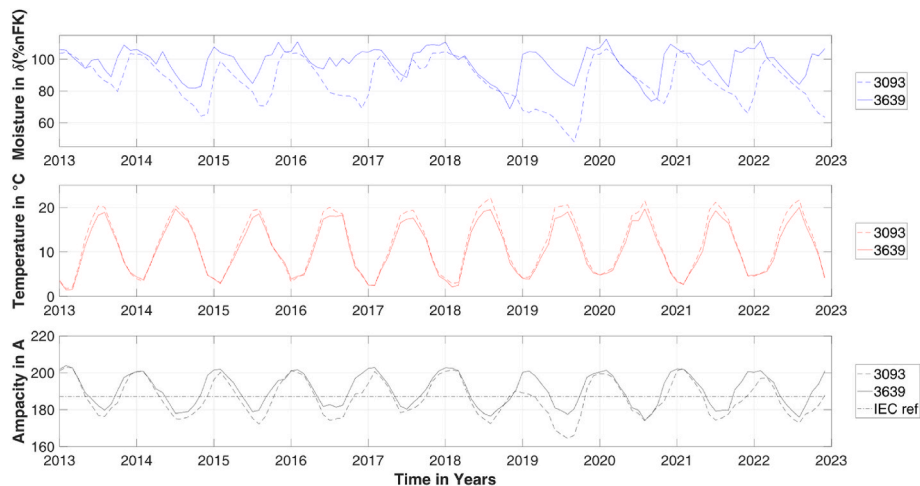


Fig. 3. Seasonal changes of monitoring stations 3093 (coastal proximity) and 3639 (inland) over 10 years.

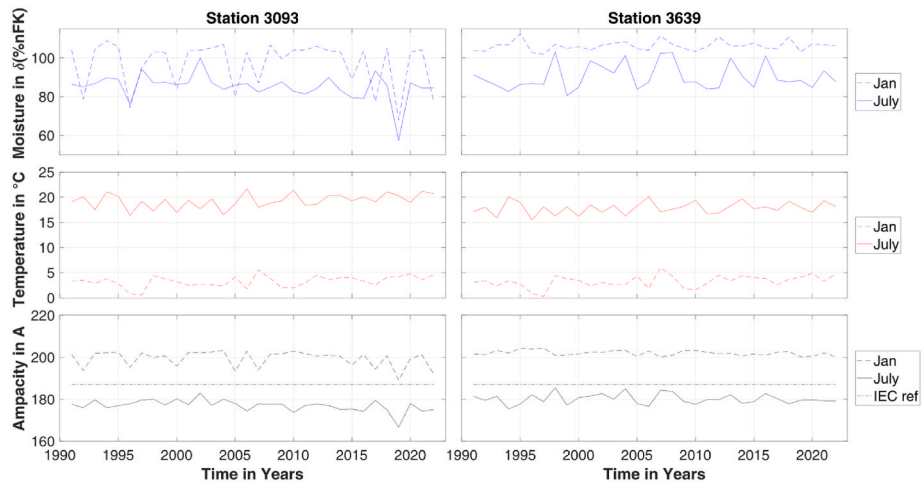


Fig. 4. Soil moisture, temperature, and ampacity of monitoring stations 3093 (left) and 3639 (right).

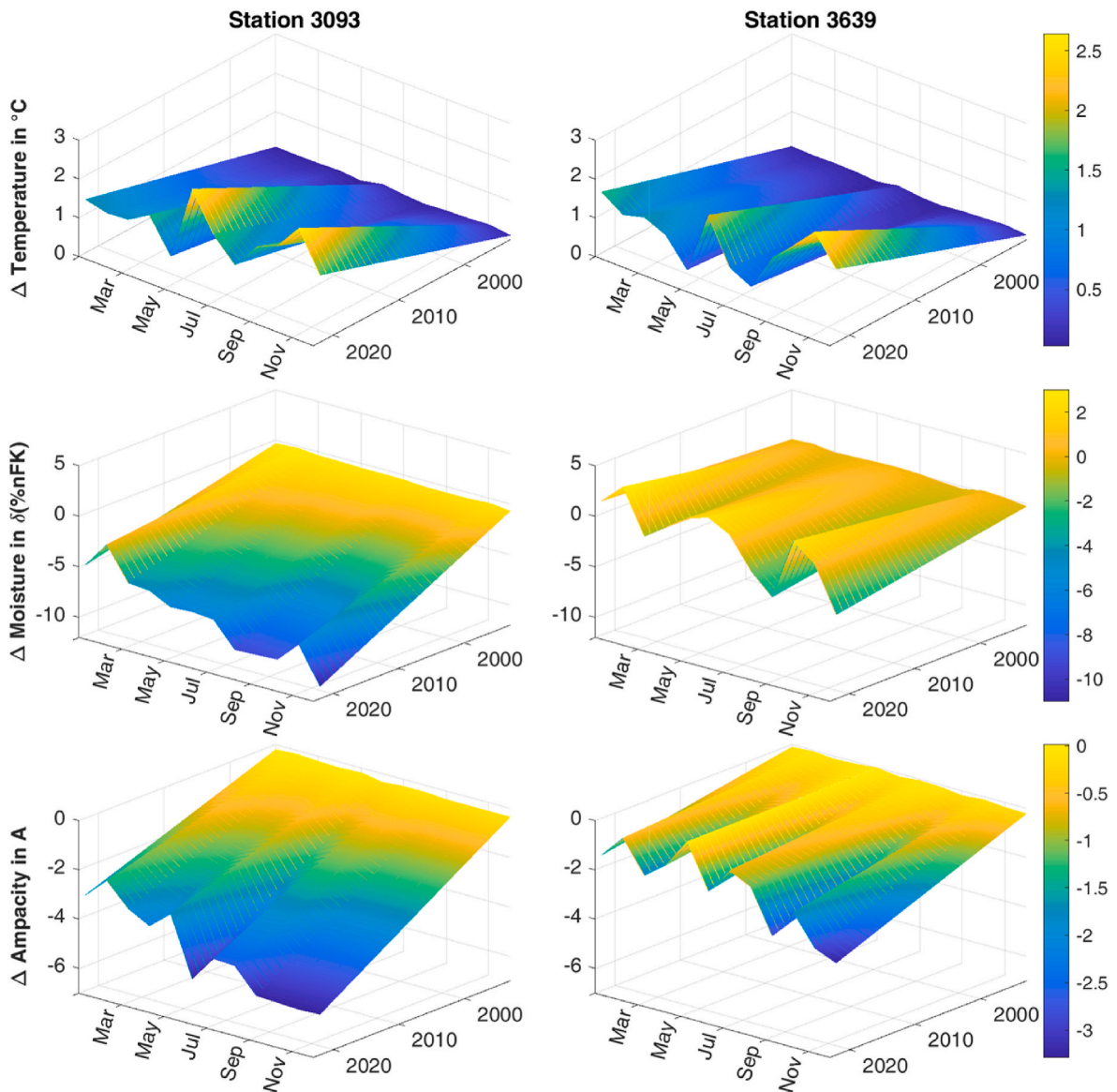


Fig. 5. Regression of monthly temperature, soil moisture, and ampacity trends relative to 1991 for measuring stations 3093 and 3639.

range from approximately 40 kVA in February to 94 kVA in November over the examined timespan. Similarly, at station 3693, reductions range from around 2 kVA in May to 57 kVA in December.

The linear regression of the soil temperatures, moistures, and the derived ampacity has been derived from a MATLAB model with the polyfit function and a polynomial degree of 1 was used to calculate the coefficients for $p(x) = p_1 \cdot x + p_2$ for each individual month and station.

The general trend of increasing temperatures and decreasing soil moisture levels and derived decreasing ampacities can be observed for all stations. Even though specific values vary and the maxima and minima shift slightly between months, the absolute range of deviation of the key data remains the same. The second half of the year tends to be more impacted. The data is mostly significant but the coefficient of determination varied throughout the results. R^2 and F-statistic values were too low to implicate further assumptions. The regression model does not indicate a future trend with the given parameters. The dataset was used nonetheless because no other data was available, and the general trend of the last 32 years is mathematically correctly represented.

5. Discussion and conclusion

The research presents a comprehensive review of the IEC standard method for calculating the ampacity of energy cables with a particular focus on the influence of seasonal and climate-related changes. To illustrate the application of the method, the metropolitan area of Hamburg was selected as a case study. The study commenced with a detailed analysis of the region's soil composition and ground characteristics, contextualizing the available weather and ground data from the German Meteorological Service (DWD). The metropolitan area and the existing DWD measuring stations were delineated, allowing for an exploration of the region's susceptibility to relevant seasonal and climate changes.

Subsequently, a thorough analysis of the IEC method was conducted, specifically identifying and describing its climate-related parameters. By performing a sensitivity analysis on local weather and soil data collected from the measuring stations, improvements were made to the ampacity calculation within the IEC method. The results were then compared against the reference values provided by the IEC, which do not account for variable climate-related parameters. Noteworthy seasonal effects and trends observed over the 32-year analysis period were demonstrated through two selected locations.

The error assessment considered the accuracy of soil data from the DWD, deviations in the IEC model, the plausibility of assumptions, and the significance of the regression model. As specific values were not critical to the study's conclusions, small error tolerances in the data, the IEC model, and the regression were disregarded. However, certain assumptions were necessary, particularly regarding the soil, its composition, pore size, and mineral content, to facilitate the derivation and comparison of the results. It was observed that the specific parameterization of these values has a substantial impact on the overall results and the comparison with the IEC standard. This variability introduces uncertainties when specific ground conditions are unknown, a frequent occurrence that further raises questions about the suitability of a generalized bottom-up approach for determining ampacity. A sensitivity analysis of the thermal resistivity itself with respect to input ground parameters was not within the scope of this contribution. The ampacity of an electric energy cable, when calculated with the IEC-method, demonstrates a negative linear correlation with the soil temperature and a negative exponential correlation with the thermal resistivity of the soil. The resistivity in turn is directly correlated with the available water content of the ground. The ampacities' sensitivity towards input changes from the standards' two extreme scenarios in Germany, resistivity of 2.5 K-m/W at a temperature of 20 °C and 1.0 K-m/W at 0 °C, is significant as it results in a total difference of almost 100 A or a relative increase of over 65%. The climate related parameters have to be properly

parameterized in order to obtain valid ampacity results. This task is difficult to accomplish in reality. The soil is generally not entirely homogeneous along a cable. Different soil structures, water levels, vegetation, man-made infrastructure, and external heat sources are only some of many factors that make a proper parametrization challenging. But even complex models that try to account for many of these quasi-static irregularities cannot include their unpredictable changes. Specifically changes in climate-related parameters and their consequences may be especially hard to forecast. These changes are discussed further.

Due to the standards premise to be universally valid, the input parameters have to be generalized in a way that allows the method to be applied everywhere. The results have to be inherently conservative and ensure that no thermal failures of the energy cable occur. This is not the case as there were instances where the ampacity and, consequently, the thermal limit was exceeded. Moreover, applying local weather and soil data results in two notable deviations from the conservative standard reference values.

The first deviation is that the energy transfer potentials of the ampacity reference values are seasonally lower than the values with incorporated ground data. These unused potentials are location dependent and exceed 10% during the winter months. The second deviation manifests itself in the over estimations of the IEC reference value due to changes in the surrounding soil conditions. This overestimation can reach up to 12%. Such instances are more likely when the IEC standard is applied in warm, dry regions with deteriorating and volatile climate conditions, compared to the relatively consistent soil conditions in the metropolitan region of Hamburg.

Both effects are noteworthy. The unused potential in transfer capacity runs contrary to the envisaged grid expansion plans. Transfer capacities are being increased on all grid levels, with all practical expansion possibilities and an enormous budget of 27 billion euro in Germany's 10-year plan alone [60,61]. They are the foundation of the transformation to a carbon-neutral supply structure. The artificial reduction of these capacities implies a noteworthy economic loss. The overestimation might lead to even more acute problems as the thermal limits of a cable could be exceeded, which would cause accelerated degradation of the cables' components which in turn shortens the lifetime or causes immediate failures. The failure of major energy cables can result in a reduction in supply security or even in direct grid overloads. The impact of a resulting blackout depends on the absolute energy capacity of the cable, the grid structure, and the current load situation. Consequences are dire in each case and do also entail cost- and time-consuming repair and maintenance work. Both effects hamper the UN's sustainable development goals, as they increase the costs associated with integrating renewable energy into the grid.

Periodic changes also occur over different time periods, e.g. the day and night cycle, correlations with regional meteorological events, or dependencies of external structural influences such as heating pipes and their usage. These changes influence the ampacity and, therefore, offer additional energy transfer potentials and pose the risk for thermal overloads. They were not considered in this article.

Superimposed onto regular seasonal changes are climate change-related variations in the environment. These will influence the planning process of electric energy cables. As an operating resource with a regular lifecycle time of over 40 years and a real usage time of often over 60 years [5], the forecast has to try to include these changing surrounding conditions. An energy transfer capacity calculated on today's situation may not be applicable over a cable's lifespan. Taking climate change-induced effects on ampacity into account, the proper planning process must not only include the varying grid demands of future periods but also the varying climate-related transfer capacities.

Beyond the general change in the environmental conditions, climate change generally causes more volatile surroundings which result in increasing changes in the transfer capacity and, more importantly, more frequent and increasingly extreme variations.

5.1. Methodological disadvantages of the planning process currently used

This subsection discusses the methodology of the rigid bottom-up approach of the IEC standard [8]. The idea to identify relevant external and internal parameters of a physical system, weigh them, and structure them in an analytical relation is a well-established procedure. Measured changes in the parameters can be used to forecast their results with the created correlation. This leads to a controllable and predictable system. That approach, however, reaches its limitations when the investigated environment becomes too complex, the input parameters become too volatile, or cannot be properly determined.

The input parameters can be accurately determined for stable environments. While periodic seasonal changes do not invalidate the approach, they increase untapped potentials that could be further utilized. This condition is likely to change due to increasingly impactful climate change. Climate-related parameters will become too volatile to be considered static and their changes will start to affect the results. Therefore, the premise of the entire approach has to be reevaluated.

5.2. Analogy to the temperature monitoring of transmission lines

Direct temperature measurement provides an alternative approach to the IEC method. Given that the underlying temperature of the conductor and the dielectric medium are significant factors limiting energy transfer capacity, alternative methods can be established by directly monitoring these temperatures and enforcing restrictions through control loops.

An operational example is the installation of temperature sensors on transmission masts that connect wind parks to the grid. The reason is that wind park energy output increases exponentially with wind speed resulting in a higher transmission load. Interestingly, higher wind speeds also enhance natural cooling through a stronger air stream around the transmission line. This creates a synergistic effect leading to mutually reinforcing increases in energy generation and transfer capacity [62]. Temperature monitoring remains valuable even without this additional correlation. By sensing and evaluating the distributed temperature, it becomes possible to directly ensure thermal stability and prevent failures. Unfortunately, this method is not directly applicable to underground cables due to limited accessibility. The conductor is not easily accessible, and the high costs associated with installation, data transfer, and the number of required sensors, combined with the inability to monitor sections instead of individual spots, make this approach impractical. It becomes evident that an alternative and suitable solution for underground cables is needed.

5.3. Criteria for practical methods to improve the determination of an energy cable's ampacity

The need for complex system models combined with the necessity for the parameterization of volatile environmental conditions in order to average out or homogenize originally heterogeneous situations and complex structures would be eliminated if thermal failures could be prevented by simply measuring the distributed temperature and therefore not relying on an arbitrary reduction of the operating current.

An essential challenge in this discussion will be the effort, economical and organisational, a dynamic line rating method will impose. Practicality has led to a static approach with significant securities in order to provide a low cost low effort solution to the actual ampacity [8, 16]. The currently employed dynamic methods usually require either additional hardware along the cable, such as fiber optic strings in the cable screen, or the cable to be disconnected from the grid, or both [63].

A practical method to maximize the energy transfer capacity and to prevent thermal failures needs to abide by the following criteria:

- **Operational usage:** The temperature has to be monitored during the operation itself. The need for disconnections or interruptions would limit the range of application.
- **Accuracy:** At least the critical temperatures e.g., of the conductor or the increase of the soil temperature, need to be accurately obtainable in order to feed control structures properly and enable them to decrease the operating current or shut the system off.
- **Protection Criteria:** The ampacity of an energy cable is a critical factor for protection equipment, as the trigger current of protective devices is typically set slightly above this threshold. Any new methodologies that enable a dynamic ampacity limit must also ensure proper adjustment of the trigger current.
- **Cable independence:** The method should work without additional distributed hardware along an energy cable. Otherwise, new installations would become more costly and old structures could not be adapted accordingly.
- **Reliability:** The system's thermal integrity depends on the proper functioning of the method. Redundant security measures and self-diagnosis-systems should be employed.
- **Hardware availability:** Measuring systems should be available and ideally commercially ready for economic, large-scale integration.
- **Flexibility and mobility:** Systems that can be added, disconnected, and shifted on demand serve a broader use case than fixed installations. Their mobility and transfer costs broaden their spectrum of usage.
- **Economic feasibility:** The cost savings need to outweigh the costs to develop, acquire, and operate a new method. The potential savings are substantial if an increase in energy transfer capacity can reduce the need to expand existing grid structures or delay the process. The total life cycle cost of a method contributes to its range of applications.

Especially the following criteria present significant challenges for new methodologies, which have thus far prevented their practical application:

- Determination of the cable temperature over the entire length of the cable
- Consideration of branches (T-branches) and distributions of cables
- Change in cable types, e.g., plastic and belt cables
- Universally applicable measuring method in practical operation, regardless of the cable type and voltage level

Research efforts pursuing methods or systems which fulfill these criteria will enhance the ampacity and consequently the energy transfer capacity of electric energy cables. This will reduce the need for grid expansions, enhance supply security through improved redundancies, and allow for dynamic grid expansions.

5.4. Benefits of clustering geographical features for ampacity boundaries

In this work, various measuring stations in the metropolitan region Hamburg were evaluated; however, a detailed analysis was conducted for only one distinguishing key feature: the proximity to the sea, comparing one coastal and one inland station. It was already suggested that there are additional significant criteria correlated with relevant soil data. These include, among others, the degree of urbanization, altitude, natural vegetation, and agricultural use of the topsoil.

It seems worthwhile to pursue further studies that cluster these factors using larger datasets, identify additional relevant distinguishing features, and derive statistical quantitative conclusions. This approach could yield correlation factors that enable more efficient operation of local power transmission lines.

CRediT authorship contribution statement

Patrick Möbius: Conceptualization, Methodology, Software, Formal analysis, Data curation, Writing – original draft, Writing – review & editing, Project administration. **Lars-Hendrik Michael:** Methodology, Software, Writing – original draft, Visualization. **Maik Plenz:** Conceptualization, Investigation, Writing – original draft. **Johannes Schröder:** Conceptualization, Investigation, Writing – review & editing, Visualization. **Detlef Schulz:** Conceptualization, Resources, Supervision, Funding acquisition.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

Historic Soil Data Set of DWD Measuring Stations in Germany: https://opendata.dwd.de/climate_environment/CDC/derived_germany/soil/

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